

# Measuring Watermark–Detector Interference: Controls, Score Attenuation, and Checkpoint Dependence

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**Abstract**—Prior work reports that embedding a provenance watermark can *harm* a passive synthetic-image detector inspecting the same pixels. That claim has a confound: any imperceptible perturbation shifts a detector trained on clean images, so degradation after marking need not be watermark-specific. We introduce a control the prior literature does not apply — a payload-free perturbation matched per image to the watermark’s own PSNR — and re-measure the effect on 20,000 held-out images under hypotheses fixed before the data was scored.

The control reverses the expected conclusion. On an EfficientNet-B6 detector, the degradation attributed to watermarking belongs to the control: matched-PSNR noise attenuates the detector’s log-odds evidence toward zero by  $\approx 16\%$  (slope 0.83–0.85), while watermarking leaves it essentially intact (slope 0.97–0.99). We then test whether this attenuation is a property of detection or of one checkpoint, by repeating the measurement on a second, provenance-disclosed detector matched to the first on baseline AUC on the same 8,000 images. The effect does not generalise to the second detector tested, and the ordering of the two arms reverses, which establishes checkpoint dependence rather than a universal property of passive detection. Matched baseline AUC closes off a weak-detector explanation; two detectors do not characterise the population of passive detectors, and we claim no more than the dependence itself.

What generalises past this dataset is analytic rather than empirical: monotone attenuation leaves AUC exactly unchanged while still altering the score-to-posterior mapping that calibration depends on, and a mean-shift statistic cannot distinguish attenuation from translation where a slope estimate can. Three supporting measurements are reported as such: both tested watermark payloads survive a face-replacement-scale splice (bit error rate 0.0000 and  $\leq 0.011$  at 10–25% coverage), so neither supplies usable manipulation evidence at that coverage; attenuation degrades isotonic calibration without reliably changing decisions under the cost model tested; and a detector-floor screen found five of six public checkpoints at chance, one advertising 98.70% accuracy scoring AUC = 0.54 in use.

**Index Terms**—digital watermarking, deepfake detection, image forensics, provenance, score calibration, AUC, checkpoint generalisation, adversarial robustness, pre-registration

## I. INTRODUCTION

Two defences against synthetic media are converging in deployment. Proactive provenance marking embeds a signal at creation time so origin can be verified later; passive detection infers manipulation from residual synthesis artefacts. Both are computed from the same pixels, so they are not independent: Wu et al. [1] report that watermarking models built for genuine images can *harm* a deployed detector, because watermark

signals overlap the forgery signals detection relies on; the same pathology is reported independently in audio [2]. If that holds generally, a detector’s score distribution shifting on marked media invalidates any decision threshold calibrated on unmarked media.

That argument has a load-bearing confound: a detector trained on clean images responds to *any* imperceptible perturbation, so degradation after marking need not implicate the watermark’s structure. No prior work in this space separates the two. We do, with a payload-free perturbation matched per image to the watermark’s own PSNR, reporting every interference quantity net of that arm, with a detector-floor precondition and a gate threshold fixed before scoring.

Our pre-registered gate returned a null on ranking, and auditing why turned the result over. The watermark contributed almost nothing; the *control* carried the effect, and carried it not as a shift but as an attenuation of the detector’s evidence toward zero. Attenuation with near-zero intercept is monotone, hence invisible to AUC by construction, which is exactly how a ranking gate returns a null while a real effect is present.

That corrected mechanism then failed its own replication. Every number above came from one third-party checkpoint with undisclosed training data, which makes “attenuation is a property of detection” and “attenuation is a property of this model” observationally identical. We built a second detector with disclosed provenance, a frozen ImageNet ResNet-18 with a logistic head fitted only on the training split, and repeated the measurement on images where the two are matched to within 0.003 baseline AUC. The ordering did not hold: on the second detector the payload-free arm attenuates *less* than the watermarked one (§V-C). The attenuation is therefore checkpoint-dependent across the two models tested, not an established property of passive detection. Two detectors cannot say which detectors exhibit it; they can say that a single-checkpoint design, run without a control arm and read through AUC, would have missed both confounds.

This paper’s spine is four results in sequence: a control that overturns an attribution, the structural reason an AUC-based read misses it, a replication that bounds how far the mechanism generalises, and what survives that bounding. Three supporting measurements are reported as such, not as independent claims of equal weight.

### 1) A control that changes the sign of the conclusion.

Every interference quantity is reported net of a payload-free perturbation matched per image to the watermark’s

own PSNR. Without it, our own headline number would have read as watermark-induced degradation that in fact belongs to the control.

- 2) **The right estimand is a slope.** A mean-shift statistic cannot distinguish attenuation from translation; attenuation with near-zero intercept is *exactly* invisible to AUC because it is monotone (verified by unit test), which is why a ranking gate can return a null while a real effect is present.
- 3) **A replication that bounds the claim** (§V-C). On EfficientNet-B6, the matched-PSNR null attenuates evidence by  $\approx 16\%$  while watermarking does not; on a second, provenance-disclosed detector matched to within 0.003 baseline AUC on the same images, the effect does not reproduce and the ordering of the two arms reverses. Two detectors establish checkpoint-dependence, not which population of detectors exhibits the effect, and we claim no more.
- 4) **Supporting measurements**, reported as support rather than as further claims. Both tested watermarks retain their payload through a face-replacement-scale splice (BER 0.0000 and  $\leq 0.011$  at 10–25% coverage, §V-D); attenuation degrades isotonic ECE 4–5 $\times$  without reliably moving decisions under the cost model tested (§V-E); and a floor screen found five of six public checkpoints at chance, one advertising 98.70% accuracy (§V-A).

## II. RELATED WORK

**Proactive watermarking.** A substantial line of work embeds a mark at creation time to establish provenance, increasingly unifying provenance with detection or localisation [3], [4]. Our concern is orthogonal: what embedding a mark does to a passive detector inspecting the same pixels, treating watermark strength (one hand-crafted, one learned scheme) as an experimental axis rather than proposing a new one.

A larger line of work is organised around the tension we measure one instance of in §V-D: robustness to benign processing and sensitivity to malicious editing pull in opposite directions, so a mark built to survive re-encoding is not thereby a good answer to “was this edited”. Semi-fragile and separable-decoder designs target localisation explicitly rather than treating it as a by-product of robustness [5], [6], [7], [8], [9], [10], [11], [12]; a separate failure surface, diffusion-based regeneration, degrades a payload globally rather than locally [13], [14], [15]. Our splice measurement probes the local case for two specific schemes and we do not generalise from it to either literature.

**Passive detection.** Reported accuracy transfers poorly once inputs are perturbed [16]; watermark robustness has been benchmarked in the same spirit [17]. Our E0 result is a sharp instance: a detector advertising 98.70% accuracy operates at chance on a distribution it was not tuned for.

**Interference.** The premise this work tests is Wu et al.’s [1] argument that watermarks devised for genuine images “may harm the deployed Deepfake detectors ... since the watermarks are prone to overlap with the forgery signals used for

detection”, answered with AdvMark, fine-tuning the watermark itself. The same pathology is reported independently in audio [2]. Guo et al. [18] benchmark passive against watermark-based detection head-to-head but never combine them, so whether marking degrades the passive detectors being benchmarked is not asked. Two gaps follow, which this paper addresses: no prior interference result controls for generic perturbation sensitivity, and none is tested under a pre-registered null. Consequences for a fused decision rule remain open; our corpus cannot settle that (Limitations).

## III. METHOD

### A. Provenance evidence as a likelihood ratio

Let the recovered payload contain  $k_e$  bit errors out of  $L$ . Under intact provenance, errors are channel noise,  $k_e \sim \text{Bin}(L, p_0)$ ; under no valid provenance the recovered bits are uninformative,  $k_e \sim \text{Bin}(L, 1/2)$ . The provenance log-likelihood ratio is

$$z_P = L \log 2 + k_e \log p_0 + (L - k_e) \log(1 - p_0), \quad (1)$$

with  $p_0$  estimated on a held-out clean-transform calibration set.  $z_P$  is in log-odds units and is defined only when a mark is present.

### B. Interference and the evidence-transfer estimator

For watermark scheme  $s$  and class  $y$ ,

$$\Delta_\mu(s, y) = \mathbb{E}[V \mid W=1, S=s, Y=y] - \mathbb{E}[V \mid W=0, Y=y], \quad (2)$$

$$\Delta_{\text{AUC}}(s) = \text{AUC}(V; W=1, S=s) - \text{AUC}(V; W=0). \quad (3)$$

The attenuation analyses of §V-B–V-C add one estimator: for each arm  $s$  we fit  $v_s = a_s + b_s v_{\text{clean}}$  by OLS, *per arm and per detector separately*, since the question is whether an arm’s slope differs from 1, not whether arms differ from each other by assumption. Every fit is *paired* on the same image. Confidence intervals on  $b_s$  and on slope differences are 95% percentile bootstrap over images (resampling an image moves all its arms together), 800–2,000 resamples depending on the analysis. Where two detectors’ slopes are compared (§V-C), both fits use the identical image set, so a difference in  $b_s$  cannot be an artefact of a difference in sample.

### C. Fusion and abstention

$$\hat{y} = \sigma(\beta_0 + \beta_1 z_P + \beta_2 V + \beta_3 W + \beta_4 (W \cdot V)). \quad (4)$$

The independence assumption implicit in naive fusion forces  $\beta_4 = 0$ ;  $\beta_4 \neq 0$  measures interference inside the decision rule. With costs  $c_{FN}, c_{FP}, c_R$ , Chow’s rule [19] gives

$$\tau_{lo} = \frac{c_R}{c_{FN}}, \quad \tau_{hi} = 1 - \frac{c_R}{c_{FP}}, \quad (5)$$

valid iff  $c_R/c_{FN} + c_R/c_{FP} < 1$ , and optimal only if  $\hat{y}$  is calibrated — the dependency this paper is about.

#### IV. EXPERIMENTAL SETUP

**Data.** A public corpus of StyleGAN-generated faces against real photographs, balanced by class: whole-image synthesis, *not* face-swap or reenactment, hence “synthetic-image detection” throughout (Limitations states the cost of that substitution). The exploratory pass used 150+150 images; the confirmatory analysis scores 20,000 held-out images across five arms (clean, two schemes, two matched nulls), and §V-C adds a second detector over 8,000 of them — 140,000 detector evaluations total, released as raw per-image scores. Splits are stratified at the image level (no identity metadata), so confidence intervals are sample-level bootstraps.

**Watermarks.** Two schemes spanning the hand-crafted/learned divide, pretrained: DWT-DCT-SVD (transform-domain) and RivaGAN (learned encoder/decoder), both at a 32-bit payload. Measured imperceptibility: 36.9 dB / 0.967 SSIM and 40.2 dB / 0.983 respectively. For every image and scheme we additionally generate a payload-free perturbation binary-searched to match *that image’s* PSNR to within 0.5 dB; all interference quantities are net of this arm. We call it the control, the null arm, or the matched-PSNR null interchangeably. Its limit should be stated plainly: PSNR matching is not spectral or perceptual matching, so the arm bounds generic perturbation sensitivity but does not fully isolate every structural difference between noise and a watermark. The defensible reading of our result is that the degradation attributed to watermarking is *also* produced by a payload-free perturbation of equal per-image PSNR, not that no watermark-specific interference of any kind exists.

**Detectors.** Six public checkpoints were screened; one, an EfficientNet-B6, clears the floor of §V-A, used zero-shot with raw logit difference  $V = \ell_{\text{fake}} - \ell_{\text{real}}$  rather than softmax confidence. Its training data is undisclosed, so results are within-model, within-image deltas. Because one undisclosed checkpoint cannot separate a property of detection from a property of a model, we add a second detector we train ourselves: a frozen ImageNet ResNet-18 feature extractor (512-d) with an  $\ell_2$ -regularised logistic head fitted on the training split alone. Validation and test are untouched by construction, and the head is deliberately linear — the most architecturally distant thing that still clears the floor, so a shared effect is evidence of generality and a divergent one is evidence against it.

**Protocol.** The detector floor, the gate threshold on  $\Delta\text{AUC}_{\text{net}}$ , and the fallback framing for a null were fixed before scoring. Every number below is produced by released code from a single configuration file.

#### V. RESULTS

##### A. Detector floor check (E0)

A detector near chance makes  $\Delta\text{AUC}$  a measurement of noise. We screened six public checkpoints; **five of the six operate at chance on our evaluation distribution** (Table I):

Two of the screened checkpoints agree to four decimal places across the same 300 images, which is consistent with one being a re-upload or a closely related checkpoint; we did not verify their provenance, so we report the coincidence rather

TABLE I  
BASELINE AUC OF EVERY SCREENED CHECKPOINT ON OUR EVALUATION DISTRIBUTION. <sup>†</sup>MODEL CARD ADVERTISES 98.70% ACCURACY.

| Checkpoint                 | Baseline AUC | Status       |
|----------------------------|--------------|--------------|
| EfficientNet-B6 (Skull1ly) | 0.8952       | clears floor |
| ViT (Wvolf) <sup>†</sup>   | 0.5365       | chance       |
| DaMsTaR/Detecto            | 0.5365       | chance       |
| prithivMLmods/v2           | 0.5299       | chance       |
| dima806                    | 0.5296       | chance       |
| Hemg                       | 0.4576       | below chance |

TABLE II  
CONFIRMATORY PRE-REGISTERED TESTS, NET OF THE CONTROL ARM, WITH 95% PERCENTILE BOOTSTRAP INTERVALS.

| Scheme      | $\Delta\mu_{\text{net}}$ (H1) | $\Delta\text{AUC}_{\text{net}}$ (H2) |
|-------------|-------------------------------|--------------------------------------|
| DWT-DCT-SVD | +0.837 [.807, .867]           | +0.0049 [.0032, .0066]               |
| RivaGAN     | +1.353 [1.317, 1.387]         | −0.0051 [−.0071, −.0033]             |

than a conclusion about it. Among the six we screened, then, the number of independent usable detectors is smaller than the count suggests. The gap between a model card’s 98.70% and  $\text{AUC} = 0.54$  on our distribution is not a rounding error, and a floor check is what stands between such a checkpoint and a table of confident noise. This screen is a precondition for our own measurements on six checkpoints, not an audit of public detectors generally. It also bounds the study: only one public checkpoint cleared the floor, which is why we trained a second, disclosed detector rather than recording the constraint as a limitation (§V-C).

##### B. Confirmatory result ( $n = 20,000$ )

Chronology, as recorded in a git-timestamped pre-registration, with each step labelled by its evidential status: (1) *exploratory* — a pass at  $n=300$  on *test* found no effect on ranking; (2) *post-hoc discovery* — repairing an unrelated defect exposed  $\Delta\mu_{\text{net}}$ , previously unexamined, on a 36-image slice, and it was large; (3) because that observation was post-hoc and cannot be confirmed on the data that produced it, we wrote a second pre-registration committing H1/H2/H3, their thresholds and their analysis code before scoring anything further; (4) *confirmatory* — the pre-registered hypotheses were then tested on *validation* ( $n=20,000$ ), which steps (1) and (2) never touched. One *disclosed deviation* occurred inside step (4): at 2% of the run a pipeline-verification script incidentally exposed interim values consistent with H1/H2. The run was not stopped, extended or otherwise altered in response, no threshold was changed afterwards, and the full deviation log is released with the code, so the ordering can be checked against git history rather than taken on trust.

Both pre-registered tests pass (Table II), but  $\Delta\mu_{\text{net}}$  is the **wrong estimand**: an audit shows the watermark contributes almost nothing for DWT-DCT-SVD (−0.009 watermark—clean vs. −0.846 null—clean; net +0.837 is the control falling). Nor is it a location shift — it ranges −0.36 to +2.28 across clean-score quintiles. Regressing each arm on the clean score gives Table III:

TABLE III  
EVIDENCE-TRANSFER REGRESSION, EFFICIENTNET-B6.

| Arm                 | slope $b$    | intercept $a$ |
|---------------------|--------------|---------------|
| DWT-DCT-SVD         | 0.968        | +0.125        |
| null (PSNR-matched) | <b>0.834</b> | −0.158        |
| RivaGAN             | 0.990        | +0.649        |
| null (PSNR-matched) | <b>0.850</b> | −0.125        |

TABLE IV  
CROSS-DETECTOR SLOPE COMPARISON, IDENTICAL IMAGES,  
800-RESAMPLE BOOTSTRAP. EFFNET-B6 = EFFICIENTNET-B6; DWT =  
DWT-DCT-SVD; RIVA = RIVA-GAN. A SLOPE BELOW 1 INDICATES  
ATTENUATION, SO A LOWER  $b$  IS MORE ATTENUATION.

| Detector  | Scheme | wm $b$ [95% CI]    | null $b$ [95% CI]         | null−wm |
|-----------|--------|--------------------|---------------------------|---------|
| EffNet-B6 | DWT    | 0.966 [.962, .971] | <b>0.832</b> [.825, .839] | −.134   |
| EffNet-B6 | Riva   | 0.987 [.983, .992] | <b>0.848</b> [.841, .856] | −.139   |
| ResNet-18 | DWT    | 0.956 [.951, .962] | 0.971 [.967, .974]        | + .014  |
| ResNet-18 | Riva   | 0.949 [.943, .954] | 0.976 [.972, .979]        | + .027  |

**The mechanism is attenuation:** the matched-PSNR null shrinks this detector’s log-odds evidence toward zero by  $\approx 16\%$  ( $b \approx 0.84$ ,  $a \approx 0$ ) while watermarking leaves it largely intact ( $b \approx 0.97$ – $0.99$ ). This is an empirical statement about EfficientNet-B6, and §V-C shows it does not reproduce on the second detector. The two consequences below are of a different kind: they follow from the statistics themselves and hold regardless of which detector is used. First, **attenuation is invisible to AUC by construction** (monotone rescaling, verified by unit test), which is why our AUC gate returned a null while a real effect was present. Second, **attenuation is precisely what breaks calibration**, shrinking log-odds toward zero and changing the score-to-posterior mapping. Effect sizes (clean-score sd 7.04): net effects of 0.12–0.19 sd, paired Cohen’s  $d_z$  of 0.38/0.55 — moderate, and class-dependent: the null arm erodes evidence on manipulated images roughly  $3\times$  more than on real ones.

### C. Replication on a provenance-disclosed detector

One undisclosed checkpoint cannot separate “attenuation is a property of detection” from “attenuation is a property of this model”. We built a detector with disclosed provenance — frozen ImageNet ResNet-18, logistic head on *training* alone — and refit the regression on the same 8,000 images restricted to the identical set both detectors scored (Table IV, Fig. 1):

Read the slope as follows:  $b = 1$  is no change relative to the clean score,  $b < 1$  is attenuation toward zero, and a *lower*  $b$  is *more* attenuation. **The ordering reverses between detectors.** On EfficientNet-B6 the matched-PSNR null sits far below the watermark arm ( $b = 0.832$  vs. 0.966 for DWT-DCT-SVD; 0.848 vs. 0.987 for RivaGAN): the null attenuates substantially more. On the provenance-disclosed detector the null sits slightly *above* the watermark arm (0.971 vs. 0.956; 0.976 vs. 0.949), so it attenuates marginally less. The intervals are tight and disjoint in both directions, so the reversal is not a resolution artefact.

The obvious objection, that a linear probe is simply too weak to express the effect, is closed by matched baseline AUC

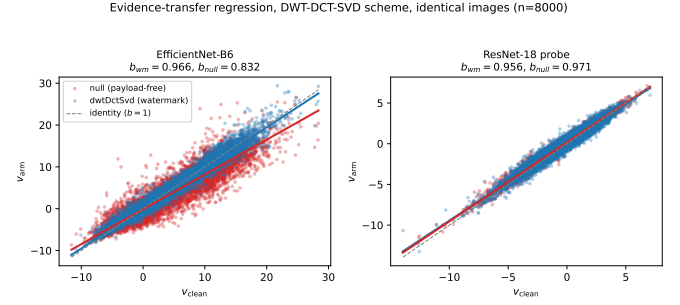


Fig. 1. Evidence-transfer regression on the same 8,000 images, DWT-DCT-SVD scheme. EfficientNet-B6 (left) shows the null arm (red) pulled below identity while the watermark (blue) tracks it; the ResNet-18 probe (right) shows the opposite ordering.

on the same images: 0.8703 (EfficientNet-B6) against 0.8725 (ResNet-18 probe), moving near-identically under every arm. The two are matched in discriminative power, not in architecture, and we claim only the former. What differs between them is how the score *distribution* moves, which is what the regression measures and AUC cannot see.

We therefore withdraw “imperceptible noise attenuates detector evidence while watermarking does not” as a claim about passive detection, and retain it only as a description of EfficientNet-B6. Why that checkpoint behaves this way is not settled here; training augmentation and its  $528\times 528$  processing resolution against the probe’s  $224\times 224$  are both plausible and neither is tested. Two consequences survive. The control arm is more strongly motivated than before, not less: without it a study reports the null’s  $-0.846$  shift as the watermark’s, and with it but only one checkpoint it reports a model-specific artefact as a property of detection. And the two structural results — that  $\Delta\mu$  conflates attenuation with translation, and that AUC is exactly blind to monotone attenuation — are analytic. They constrain what any such experiment can detect, and no replication outcome, ours included, bears on them.

### D. Payload survival under localised manipulation

The source framework routes “is the watermark present, and has it been altered?” into detection. We supplied the missing manipulation step — watermarking an authentic image, *then* splicing a donor region in with Poisson blending — and measured BER against manipulated area, for both schemes, at six coverage levels (Fig. 2).

**What was measured.** At 10–25% coverage, the range a face replacement occupies, DWT-DCT-SVD recovers its payload exactly (BER 0.0000) and RivaGAN recovers it to within 0.0052–0.0104, against a payload-destruction reference of 0.5. BER departs from zero only well beyond that range — still 0.0026 (DWT-DCT-SVD) at 40% coverage, and 0.0443/0.0521 at 60% — and even obliterating 95% of the image leaves DWT-DCT-SVD at 0.2630, short of the destruction reference. The payload is therefore not uninformative everywhere; it is uninformative over the coverage range that matters for face replacement, and only weakly informative well outside it.

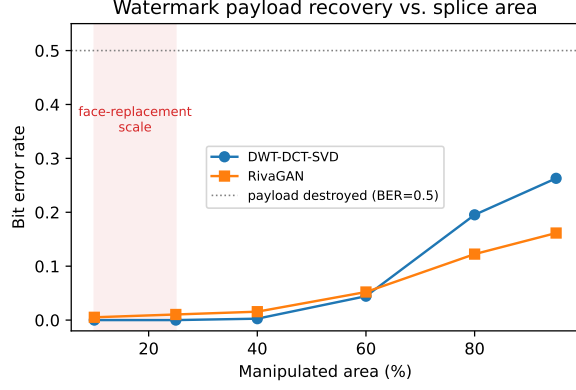


Fig. 2. BER of the recovered payload vs. spliced area. The shaded band is realistic face-replacement coverage; both curves are near-zero across it.

TABLE V  
EXPECTED CALIBRATION ERROR BY ARM AND CALIBRATOR.

| Arm           | Platt ECE | Isotonic ECE  | slope vs. clean |
|---------------|-----------|---------------|-----------------|
| clean         | 0.1273    | <b>0.0094</b> | 1.000           |
| DWT-DCT-SVD   | 0.1228    | 0.0154        | 0.968           |
| RivaGAN       | 0.1353    | 0.0290        | 0.990           |
| null[DWT]     | 0.0958    | <b>0.0492</b> | 0.834           |
| null[RivaGAN] | 0.1009    | <b>0.0456</b> | 0.850           |

*What it suggests.* The pattern is consistent with the spatial redundancy expected of a robust scheme, rather than with a peculiarity of this corpus: the redundancy that lets a mark survive JPEG, resizing and re-encoding is the same redundancy that lets it survive a region being overwritten.

*What is not established.* We tested two schemes and one splice mechanism at six coverage levels. That is not a class-level result about robust watermarking, and we do not state it as one; a differently constructed robust scheme, or a manipulation that disturbs the carrier globally rather than locally, could behave differently. Within the tested configuration the consequence for fusion is concrete: the provenance feature is effectively constant across the manipulation levels of interest, so a fused rule over it combines one informative signal with one that does not vary. This is also the design pressure behind the semi-fragile and separable-decoder schemes of §II.

#### E. Calibration under attenuation

Attenuation degrades calibration, not just rank; we measure this with ECE under Platt and isotonic calibration [20], fit on the *clean* half of a calibration split and frozen before evaluation (Table V, Fig. 3):

The two calibrators disagree, and we report the disagreement rather than selecting the one that suits the argument. Under isotonic regression, attenuation degrades ECE 4–5 $\times$  relative to clean while watermarking degrades it far less. Under Platt scaling the null arms appear to *improve*. The disagreement resolves on evidence independent of the comparison: Platt’s ECE on the *clean* arm is already 13 $\times$  isotonic’s (raw logits span  $-11.6$  to  $+36.8$ ,  $sd = 7.04$ ), so a single logistic is badly misspecified for these scores before any perturbation

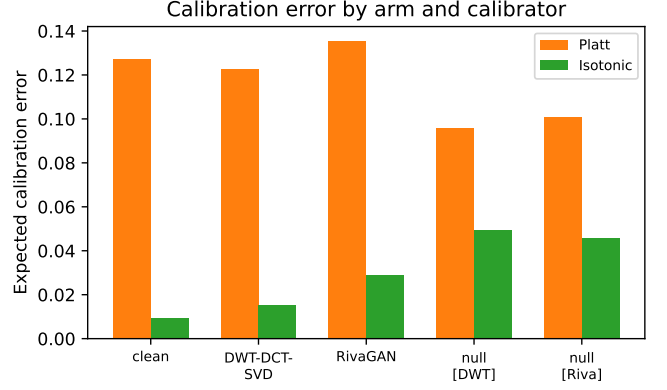


Fig. 3. ECE by arm under each calibrator. Isotonic shows attenuation degrading calibration; Platt shows the opposite, resolved by its badly misspecified clean-arm baseline (0.1273, 13 $\times$  isotonic’s).

is applied, and shrinking them happens to partially cancel that misspecification. A  $\Delta$ ECE measured against a broken baseline measures the baseline.

The claim therefore holds for calibration *quality* and does not hold for decision *risk*: isotonic DRD is 0.0055 (clean) against 0.0145/0.0054 (nulls), small and inconsistent in direction. Attenuation reaches the probability estimates without reliably changing the decisions those estimates drive *under the cost model tested here*. We make no claim that attenuation is decision-neutral under other cost structures, nor that it always causes decision harm; the cost triple is assumed rather than elicited from a deployment (Limitations).

#### F. The fusion comparison is not estimable; an exploratory transform analysis

Two distinct points, which we separate deliberately because conflating them would misrepresent the second.

First, the planned fusion comparison (F0–F5, with and without the interaction term of Eq. 4) under localised manipulation is *not estimable* in this setup. §V-D shows the provenance feature is effectively constant across the manipulation coverage of interest, and a coefficient on a feature that does not vary is not a weak estimate but an undefined one. We therefore report no confirmatory fusion result, and nothing below is offered as a substitute for it.

Second, and separately, we ran an *exploratory* analysis asking a narrower question: does the interaction term respond to benign input changes at all, on otherwise unmanipulated images? This is reported solely to characterise the term’s behaviour, not to establish any fusion claim. Parameters:  $n=400$  (120 calibration / 280 test, held-out *test* split, disjoint from validation), EfficientNet-B6, seven transforms, and *no bootstrap intervals* — point estimates only, at a sample size an order of magnitude below the confirmatory analyses. Results are in Table VI.

F0 and F4 AUC fall together under compression and down-scaling, which tracks the detector’s own fragility rather than anything watermark-specific.  $\beta_4$  stays small ( $|\beta_4| < 0.08$ ) for six of the seven transforms. Resize-to-0.5 $\times$  is the exception

TABLE VI  
TRANSFORM ROBUSTNESS, F0 (NO INTERACTION) VS. F4 (WITH IT),  
N=400.

| Transform    | F0 AUC | F0 DRD | F4 AUC | F4 DRD       | F4 $\beta_4$  |
|--------------|--------|--------|--------|--------------|---------------|
| JPEG(90)     | 0.899  | 0.006  | 0.897  | 0.008        | +0.004        |
| JPEG(70)     | 0.776  | 0.003  | 0.819  | 0.005        | -0.022        |
| JPEG(50)     | 0.721  | 0.001  | 0.794  | 0.005        | +0.038        |
| Resize(0.75) | 0.743  | 0.003  | 0.825  | 0.008        | +0.048        |
| Resize(0.5)  | 0.710  | 0.001  | 0.825  | <b>0.050</b> | <b>+0.221</b> |
| Bright(+20%) | 0.909  | 0.034  | 0.909  | 0.031        | +0.056        |
| Bright(-20%) | 0.841  | 0.003  | 0.848  | 0.006        | -0.072        |

( $\beta_4 = +0.221$ , DRD 0.050). With  $n=280$  test images and no interval, we flag that cell as the one worth a CI-backed rerun and claim nothing from it. A  $\rho$  sweep (fusion fit once on calibration, then evaluated on mixtures at watermarked fractions  $\rho \in \{0.05, 0.25, 0.5, 1.0\}$ ) found AUC, ECE and DRD essentially flat across  $\rho$  for both schemes, which is what the rest of the paper would predict: a feature that does not move the outcome at  $\rho=1$  will not start mattering because it appears less often. Full per- $\rho$  values are in the released results.

## VI. LIMITATIONS

**Scope of the manipulation result.** We supplied the manipulation step the source framework presupposes and measured the payload surviving it at realistic coverage (§V-D), so the channel’s silence is measured rather than assumed. The measurement covers two schemes and one Poisson-blended elliptical splice. It is not a class-level statement about robust watermarking, and our splice reproduces the provenance mechanics of a face swap without reproducing a learned swap’s detector-facing realism, so it does not speak to the detectability of modern face swaps.

**Statistical power.** The  $n=300$  exploratory pass has AUC standard error  $\approx 0.03$ , larger than the effects measured, so its null means “undetectable at this size”. Confirmatory and replication analyses ( $n=20,000/8,000$ ) do not have this problem; reported intervals, not point estimates, bound excluded effect sizes.

**Synthesis type.** Whole-image GAN synthesis vs. real photographs, not face-swap/reenactment as in standard benchmarks [21]; detectors keying on generator fingerprints need not generalise to different manipulation traces.

**Detector provenance and count.** The strong detector’s training data is undisclosed (mitigated by within-model, within-image deltas); the disclosed detector is a linear probe on generic features, not an engineered detector. Two detectors establish non-universality, not which population of detectors exhibits the effect — that needs a panel, and §V-A shows that floor-clearing detectors are the scarce resource.

**Other.** Interference is measured zero-shot; a task-trained detector is untested. Decision thresholds use an assumed cost triple and swept, not measured,  $\rho$ .

## VII. DISCUSSION

**What a null does and does not mean.** We did not detect [1]’s effect in our setting, which is a boundary condition, not

a refutation: their detectors are trained for the manipulation family they evaluate and their images are manipulated *after* marking, so watermark and forgery signals coexist as their argument describes. Ours are zero-shot detectors on whole-image synthesis marked after the fact.

**Our own gate had the asymmetry it warns about.** We gated on  $\Delta\text{AUC}_{net}$ , while the quantity a calibration-centred thesis needs is  $\Delta\mu_{net}$ . Every reading of the latter is post-hoc with respect to that gate, which is why it was re-registered before being tested rather than reinterpreted in place.

**One checkpoint is not a detector.** The replication of §V-C is the result we would have preferred to avoid and the one most worth reporting. Our probe is not the right second detector in any principled sense — it is a weak linear head on generic features — but a difference measured on a single checkpoint is not yet a finding, and the cheapest available remedy, a frozen backbone with a linear head trained on a named split, costs little enough that its absence is hard to justify. The detector-floor screen serves the same precondition role: a study measuring a *difference* in detector behaviour assumes the detector behaves at all.

## VIII. CONCLUSION

We set out to measure whether provenance watermarking interferes with passive synthetic-image detection, and the defensible answer is narrower than the one we started toward. Against a payload-free control matched per image to the watermark’s PSNR, on 20,000 held-out images, an EfficientNet-B6 detector’s evidence is attenuated by  $\approx 16\%$  under the control and left largely intact under watermarking, so the degradation that setup would attribute to the watermark is not attributable to it on that checkpoint. On a second, provenance-disclosed detector matched to the first within 0.003 baseline AUC on the same images, the effect does not reproduce and the ordering of the two arms reverses. Across the two models tested, the attenuation is therefore checkpoint-dependent; which detectors exhibit it is a question two models cannot answer, and we do not answer it.

What generalises is analytic rather than empirical. A mean-shift statistic conflates attenuation with translation, and AUC is exactly blind to monotone attenuation, so a design that measures with  $\Delta\mu$  and gates on  $\Delta\text{AUC}$  can be wrong about the mechanism and unable to see it at the same time; a slope estimate is the estimand that separates them. Three supporting measurements accompany this. Both tested watermarks retain their payload through a face-replacement-scale splice, providing no detectable evidence of manipulation at that coverage in our experiment, though BER does rise at far larger coverage. Attenuation degrades isotonic calibration without reliably changing decisions under the cost model tested. And a detector-floor screen found five of six public checkpoints at chance, one advertising 98.70% accuracy, which is a precondition for trusting any of the above rather than a finding about public detectors at large.

None of this shows that watermark-detector interference is absent, and none of it shows that robust watermarks are generally unable to evidence tampering. What it shows is narrower

and more awkward: the single-checkpoint, no-control, AUC-based apparatus examined here can confuse a watermark effect with generic perturbation sensitivity, a property of one model with a property of detection, and a calibration-relevant change in the score distribution with no change at all.

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